

Answer Key

Grammar

Future Perfect: achievements by a future point in time

The Future Perfect (will have + past participle) describes an action that will be completed before a specific point in the future. It is often used with expressions like 'by 2035' or 'by the end of the mission'.

Future Perfect (will have + fiilin 3. hâli), gelecekteki belirli bir zamandan ÖNCE tamamlanmış olacak bir eylemi anlatır. 'By 2035' (2035'e kadar), 'by the end of the mission' (görevin sonuna kadar) gibi ifadelerle sık kullanılır.

When? We use it to talk about achievements, missions or changes that will be complete by a future date or event — very common when we predict the future of space exploration.

Gelecekteki bir tarihe ya da olaya kadar tamamlanmış olacak başarıları, görevleri veya değişiklikleri anlatırken kullanılır — özellikle uzay araştırmaları ve gelecek tahminlerinde.

Positive

- Scientists will have built a Moon base by 2035.
- The rover will have explored the surface by next year.

Negative

- Engineers won't have finished the new rocket by spring.
- We won't have solved every problem by then.

Question

- Will astronauts have landed on Mars by 2045?
- Will this telescope have discovered new planets by 2040?

Short answers

- Yes, they will. / No, they won't.
- Yes, it will have. / No, it won't have.

Common mistakes

- ~~By 2035, scientists will build a Moon base.~~ → **By 2035, scientists will have built a Moon base.**
- ~~Will astronauts landed on Mars by 2045?~~ → **Will astronauts have landed on Mars by 2045?**

1. By 2035, engineers _____ a permanent base on the Moon.

- a. will build
- b. will have built ✓
- c. have built

'By 2035' → *tamamlanmış olacak eylem: will have built.*

2. The crew _____ six months on the station by the end of the mission.

- a. will have spent ✓
- b. will spend
- c. spend

Görevin sonuna kadar tamamlanmış olacak: will have spent.

3. She _____ her training before the next launch.

- AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.
- a. won't finish
 - b. won't have finished ✓
 - c. doesn't finish

Olumsuz Future Perfect: won't have finished.

4. _____ humans _____ on Mars by 2045?

- a. Will / land
- b. Will / have landed ✓
- c. Do / land

Soru: Will + özne + have + 3. hâl.

5. This telescope _____ (discover) hundreds of new stars by 2040. **(will have discovered)**

Future Perfect: will have + 3. hâl.

6. Engineers _____ (not / finish) the new rocket by next spring. **(won't have finished)**

Olumsuz: won't have + 3. hâl.

7. By 2050, scientists will solve every problem of long space travel. — True / False **(Incorrect — By 2050, scientists will have solved every problem of long space travel.)**

8. The rover will have explored the surface by next year. — True / False **(Correct)**

9. A: _____ scientists _____ (build) a Mars colony by 2100? **(Will / have built)**

10. B: I think they _____ (build) the first small colony, but the big cities _____ (not / appear) yet. **(will have built / won't have appeared)**

11. Make a question about the future of space travel.

Will astronauts have visited another galaxy by 2200?

Future Perfect Progressive: how long something will have been happening

The Future Perfect Progressive (will have been + verb-ing) emphasises the DURATION of an activity that continues up to a point in the future. It focuses on how long something has been happening, not just the result.

Future Perfect Progressive (will have been + fiil-ing), gelecekteki bir noktaya kadar SÜREN bir eylemin uzunluğunu vurgular. Sonuçtan çok 'ne kadar süredir' devam ettiğine odaklanır.

When? We use it to say how long a mission, project or habit will have been continuing by a certain future time.

Bir görevin, projenin ya da alışkanlığın gelecekteki bir zamana kadar ne kadar süredir devam ediyor olacağını anlatırken kullanılır.

Positive

- The crew will have been living on the station for a year by June.
- This rover will have been exploring the surface for twelve months by then.

Negative

- Engineers won't have been testing the suit for long before the launch.
- We won't have been travelling for more than a week by that point.

Question

Short answers

- How long will the telescope have been orbiting the Earth by 2040? • Yes, they will have been. / No, they won't have been.
- Will they have been working on this project for over a decade?

Common mistakes

- ~~By June, the crew will have living on the station for a year.~~ → **By June, the crew will have been living on the station for a year.**
- ~~How long will the telescope orbiting the Earth by 2040?~~ → **How long will the telescope have been orbiting the Earth by 2040?**

1. By the end of the mission, the crew _____ in zero gravity for a whole year.
- a. will have lived
 - b. will have been living ✓
 - c. live

Süre vurgusu: will have been living.

2. Next spring, this rover _____ the surface for twelve months.
- a. will have been exploring ✓
 - b. will explore
 - c. explores

Belirli bir süre + gelecekteki nokta → Future Perfect Progressive.

3. Engineers _____ the new spacesuit for long before the launch.
- a. won't test
 - b. won't have been testing ✓
 - c. don't test

Olumsuz: won't have been + fiil-ing.

4. How long _____ (the telescope / orbit) the Earth by 2040? **(will the telescope have been orbiting)**

Soru: How long will + özne + have been + fiil-ing?

5. By 2050, scientists _____ (study) this planet for almost thirty years. **(will have been studying)**

Süreye vurgu: will have been studying.

6. By June, the crew will have been living on the station for a year. — True / False **(Correct)**

7. How long will the telescope orbiting the Earth by 2040? — True / False **(Incorrect — How long will the telescope have been orbiting the Earth by 2040?)**

8. A: How long _____ (you / train) as an astronaut by the time of the launch? **(will you have been training)**

9. B: By then, I _____ (train) for almost three years. **(will have been training)**

10. The team is building the space colony. They will continue for ten years, by 2100. *(Combine into one sentence with the Future Perfect Progressive.)* **By 2100, the team will have been building the space colony for ten years.**

Question tags with the Future Perfect

Question tags check if something is true. If the main sentence is positive (will have + past participle), the tag is negative (won't + pronoun), if the main sentence is negative (won't have + past participle), the tag is positive (will + pronoun). The auxiliary verb in the tag is always 'will'.

Question tag'ler, cümlelerin doğru olup olmadığını teyit etmek için kullanılır. Ana cümle olumluysa (will have + 3. hâl) tag olumsuzdur (won't + zamir); ana cümle olumsuzsa (won't have + 3. hâl) tag olumludur (will + zamir). Yardımcı fiil her zaman aynıdır: will.

When? We use question tags to confirm a prediction or check someone's opinion — very useful in conversations and debates about the future.

Bir tahmin veya varsayımı doğrulamak, karşıımızdaki kişinin fikrini sormak için kullanılır; özellikle gelecekle ilgili konuşma ve tartışmalarda kullanışlıdır.

Positive

- The astronauts will have landed by then, won't they?

Negative

- This telescope won't have finished its survey by 2030, will it?

Question

- You will have joined mission control by the time we launch, won't you?

Short answers

- Yes, they will. / No, they won't.

Common mistakes

- ~~The astronauts will have landed by then, don't they?~~ → **The astronauts will have landed by then, won't they?**
- ~~We won't have solved every problem by 2050, won't we?~~ → **We won't have solved every problem by 2050, will we?**

1. The astronauts will have landed by then, _____ ?

- a. won't they ✓
- b. will they
- c. don't they

Olumlu cümle → olumsuz tag.

2. This telescope won't have finished its survey by 2030, _____ ?

- a. won't it
- b. will it ✓
- c. isn't it

Olumsuz cümle → olumlu tag.

3. You will have joined mission control by the time we launch, _____ ?

- a. don't you
- b. won't you ✓
- c. aren't you

Olumlu cümle → olumsuz tag, aynı yardımcı fiil (will).

4. We won't have solved every problem of space travel by 2050, _____ ? (will we)

Olumsuz cümle → olumlu tag.

5. The rover will have been exploring the surface for a year by June, _____? (**won't it**)

Olumlu cümle → olumsuz tag. AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.

6. This telescope won't have finished its survey by 2030, will it? — *True / False* (**Correct**)

7. You will have joined mission control by the time we launch, don't you? — *True / False* (**Incorrect — You will have joined mission control by the time we launch, won't you?**)

8. A: By 2100, engineers will have built the first Mars colony, _____? (**won't they**)

9. B: I hope so! But scientists won't have solved every problem, _____? (**will they**)

10. Make a tag question about a rover exploring a planet.

The rover will have been exploring the planet for a year by then, won't it?

Life Aboard the Space Station

Before you read

- Look at the title. What do you think an astronaut's daily routine looks like?
- Guess: how many hours a day do you think astronauts spend exercising?

My name is Commander Lena Voss, and I have been living on the International Research Station for four months. By the end of this mission, I will have spent almost a year of my life in zero gravity.

Every morning starts the same way. I check my spacesuit, talk to mission control, and review the day's tasks. Without gravity, even simple things like eating breakfast or brushing your teeth become a small adventure — food floats if you're not careful!

In the afternoon, we usually run experiments. Right now, we are growing plants to see if a colony on the Moon could one day grow its own food. By the time I return to Earth, our team will have completed over two hundred experiments.

Exercise is extremely important here. Without it, our muscles and bones become weak in zero gravity, so we exercise for two hours every day. By the end of my mission, I will have been exercising on this station for almost a thousand hours.

In the evening, I like to look out of the window at Earth. It's easy to forget how small our planet looks from up here. I always remind the younger astronauts: by the time your first mission ends, you will have seen more sunrises than most people see in a lifetime — the station orbits the Earth every ninety minutes!

Life here isn't always easy, but I wouldn't change it. When people ask if it's worth it, I always say the same thing: humans will have learned enough about long space missions by the 2030s to finally send astronauts to Mars — and I want to be part of that story.

1. What is the text mainly about?

- a. The history of space stations
- b. An astronaut's daily life and future plans ✓
- c. How rockets are built

2. How long has Lena already been on the station?

- a. Four months ✓
- b. A year
- c. Two hundred days

3. Why does Lena exercise every day?

- a. To feel less bored
- b. Because her muscles and bones become weak in zero gravity ✓
- c. Because mission control tells her to compete

4. What are the astronauts growing plants for?

- AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.
- a. To decorate the station
 - b. To test food for a possible Moon colony ✓
 - c. To sell them on Earth

5. The space station orbits the Earth once a day. — True / False (**Incorrect — The space station orbits the Earth every ninety minutes.**)

6. How does Lena probably feel about her life on the station?

- a. Regretful
- b. Proud and hopeful about the future of space travel ✓
- c. Bored

After you read

- By the end of Lena's mission, what will she have done? Write two sentences with the Future Perfect.
- Would you like to live on a space station for a year? Why or why not?

Our Mars Colony Project

Before you read

- What do you think people would need to survive on Mars?
- Guess: what jobs might people have in a colony on another planet?

For our class project this term, we designed a hypothetical colony on Mars. First, we had to decide who would live there: scientists, engineers, doctors, and farmers who could grow food in a sustainable way.

We predicted that by 2100, the first colony will have grown from just a few hundred people to a small, functioning town. Our group decided the colony would need its own sustainable energy system, because sending fuel from Earth would be far too expensive.

One of the biggest challenges was breathing. Mars doesn't have enough oxygen, so every building needs a special system. We imagined that by the time the colony celebrates its fiftieth anniversary, engineers will have developed advanced technology to make the air completely safe, even outside the main buildings.

We also wrote short scripts for our presentation. In one scene, a scientist explains a new breakthrough in growing food on Mars; in another, two astronauts argue about whether humans should build more colonies or focus on protecting Earth instead.

Working on this project taught us that interplanetary travel isn't just about rockets and spacesuits — it's also about people, teamwork, and difficult decisions. We will have learned a lot more about ourselves than about Mars by the time we finish presenting our project to the class!

1. What was the class project about?

- a. Building a real spacecraft
- b. Designing a hypothetical colony on Mars ✓
- c. Writing a report about the Moon

2. Why does the colony need its own sustainable energy system?

- a. Because sending fuel from Earth would be too expensive ✓
- b. Because Mars is too cold
- c. Because the colony has no engineers

3. What is one of the biggest challenges for the colony, according to the text?

- a. Finding water
- b. Breathing, because Mars doesn't have enough oxygen ✓
- c. Building roads

4. What do the two astronauts argue about in the script?

- a. Who should be the leader
- b. Whether to build more colonies or protect Earth instead ✓
- c. What to eat for dinner

5. The students think their project was only about rockets and spacesuits. — *True / False* **(Incorrect — The students learned the project was also about people, teamwork, and difficult decisions.)**

6. What can we infer about the writer's opinion of the project?

- a. It was a waste of time
- b. It was a valuable and interesting learning experience ✓
- c. It was too easy

After you read

- What job would you choose in a Mars colony? Explain why in two sentences.
- Write one prediction about Mars colonies using the Future Perfect, e.g. 'By 2100, ...'

Planning the Next Mission

Before you listen

- You will hear two astronauts, Kaan and Zara, planning their next mission to the space station.
- Predict: what do you think astronauts have to prepare before a mission?

1. What are Kaan and Zara preparing for?

- a. A conference
- b. Their next mission to the space station ✓
- c. A job interview

2. What will Zara have completed by Friday?

- a. Her spacesuit
- b. The last zero-gravity simulation ✓
- c. Her research proposal

3. How long will they have been living on the station by the end of the mission?

- a. Four months
- b. Six months
- c. Eight months ✓

4. What is Kaan's research about?

- a. Building spacesuits
- b. How plants grow without gravity ✓
- c. Repairing telescopes

5. Kaan and Zara will not be able to talk to their families at all during the mission. — *True / False* (**Incorrect — They won't see their families, but they will have video calls.**)

Transcript (teacher use)

Kaan So, by the time we launch next month, we have to finish all our training. How is yours going?

Zara Good. By Friday, I will have completed the last zero-gravity simulation. What about the spacesuit checks?

Kaan Mission control says they will have finished testing our suits by next week.

Zara Perfect. And by the end of the mission, we will have been living on the station for eight months.

Kaan Eight months! We won't have seen our families for a very long time, will we?

Zara No, we won't. But we'll have video calls. Will you have finished your research proposal before we go?

Kaan Yes, I will have sent it to mission control by Monday. I'm studying how plants grow without gravity.

Zara That's a great project. By 2035, that research could help build the first sustainable colony!

An Interview with a Futurist

Before you listen

- You will hear a radio interview with Dr Aylin Baran, a scientist who studies future technology.
- Predict: what changes do you think a futurist might talk about?

AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key — Orijinal içerik, TYMM müfredatıyla uyumlu.

1. Who is Dr Aylin Baran?

- a. An astronaut
- b. A scientist who studies future technology ✓
- c. A radio host

2. According to Dr Baran, what will have changed almost every part of our lives by 2075?

- a. Sustainable colonies
- b. Artificial intelligence ✓
- c. Telescopes

3. What does Dr Baran say about jobs in the future?

- a. All jobs will disappear
- b. Some jobs will disappear, but new ones will appear ✓
- c. Nothing will change

4. What breakthrough is Dr Baran most excited about?

- a. Faster telescopes
- b. Clean, sustainable energy for spacecraft ✓
- c. New spacesuits

**5. Dr Baran believes scientists will only start researching sustainable colonies after 2075. — True / False
(Incorrect — She believes scientists will have been researching sustainable colonies for decades by 2075.)**

Transcript (teacher use)

Host Dr Baran, thank you for joining us. What will everyday life look like in fifty years?

Dr Baran Well, by 2075, artificial intelligence will have changed almost every part of our lives — from transport to medicine.

Host That sounds exciting, but also a little frightening. Won't people have lost their jobs?

Dr Baran Some jobs will have disappeared, yes, but new ones will have appeared too — especially in space exploration and sustainable technology.

Host Do you think we will have built a colony on Mars by then?

Dr Baran I do. Scientists will have been researching sustainable colonies for decades by 2075, so I'm optimistic.

Host What breakthrough are you personally most excited about?

Dr Baran Clean, sustainable energy for spacecraft. Once we solve that, interplanetary travel will finally become realistic for ordinary people, not just astronauts.

Revision

AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.

1. Choose the correct meaning: 'sustainable'
 - a. something that can continue without running out ✓
 - b. something dangerous
 - c. something very old
2. What does 'breakthrough' mean?
 - a. a big, important discovery ✓
 - b. a small mistake
 - c. a type of spacecraft
3. By 2035, engineers _____ a permanent base on the Moon.
 - a. will build
 - b. will have built ✓
 - c. build
4. By the end of the mission, the crew _____ in zero gravity for a whole year.
 - a. will have lived
 - b. will have been living ✓
 - c. live
5. The astronauts will have landed by then, _____ ?
 - a. won't they ✓
 - b. will they
 - c. don't they
6. This telescope won't have finished its survey by 2030, _____ it? (will)
7. In 'Life Aboard the Space Station', why does Lena exercise every day?
 - a. To feel less bored
 - b. Because her muscles and bones become weak in zero gravity ✓
 - c. Because it is required for fun
8. In 'Our Mars Colony Project', the students only learned about rockets and spacesuits. — True / False
(Incorrect — They also learned about people, teamwork, and difficult decisions.)

Theme Test

AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.

1. A person trained to travel and work in space is a(n) _____.
 - a. engineer
 - b. astronaut ✓
 - c. pilot
2. The condition where objects float because there is no weight is called _____.
 - a. zero gravity ✓
 - b. orbit
 - c. launch
3. Scientists hope this _____ will change space travel forever. (çığır açan buluş) **(breakthrough)**
4. A vehicle that explores the surface of a planet is called a _____.
 - a. telescope
 - b. rover ✓
 - c. spacesuit
5. A system that can continue for a long time without running out of resources is _____. (sürdürülebilir) **(sustainable)**
1. By 2040, this telescope _____ hundreds of new stars.
 - a. will discover
 - b. will have discovered ✓
 - c. discovers
2. By June, the crew _____ on the station for a whole year.
 - a. will have lived
 - b. will have been living ✓
 - c. live
3. Engineers _____ the new rocket by next spring.
 - a. won't have finished ✓
 - b. won't finish
 - c. don't finish
4. You will have joined mission control by the launch, _____?
 - a. don't you
 - b. won't you ✓
 - c. aren't you
5. How long _____ (the rover / explore) the surface by next spring? **(will the rover have been exploring)**
6. We won't have solved every problem of space travel by 2050. (Add a correct question tag.) **We won't have solved every problem of space travel by 2050, will we?**

1. According to 'Life Aboard the Space Station', how long will Lena have spent in zero gravity by the end of her mission? AbOnlinEnglish Upper Secondary — Lise Hazırlık — Answer Key · Orijinal içerik, TYMM müfredatıyla uyumlu.

- a. Four months
- b. Almost a year ✓
- c. Ten years

2. In 'Our Mars Colony Project', why does the colony need its own energy system?

- a. Sending fuel from Earth would be too expensive ✓
- b. Mars has no sunlight
- c. The colony has no engineers

3. The space station in 'Life Aboard the Space Station' orbits the Earth once a day. — True / False **(Incorrect — It orbits the Earth every ninety minutes.)**

1. Make a prediction with the Future Perfect about a Mars colony by 2100.

By 2100, humans will have built a colony on Mars.

2. Ask about the duration of a rover's mission using 'How long'.

How long will the rover have been exploring the surface by next year?

3. This telescope won't have finished its survey by 2030, _____ it? **(will)**

1. scientists / build / a sustainable colony / by 2100 *(Make a correct sentence with the Future Perfect.)*

Scientists will have built a sustainable colony by 2100.

2. the crew / not / finish / their research / before the next mission *(Make a correct sentence with the Future Perfect.)* **The crew won't have finished their research before the next mission.**